Maruf Adesina Lawal

Ph.D. Candidate in Mathematics | Mathematical Biology

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RESEARCH INTERESTS

Mathematical biology and ecology; infectious disease modeling; optimal control theory; natural resource modeling; epidemic forecasting; model calibration and validation against surveillance data; machine learning and AI for health.

EDUCATION

Ph.D., Mathematics, University of Tennessee, Knoxville Expected Dec. 2026
Dissertation: *Modeling COVID-19 Dynamics Incorporating Vaccine Hesitancy and An Optimal Control Framework for Multi-Species Harvesting in the Southern Black Sea*
Advisor: Prof. Suzanne Lenhart

M.S., Mathematics, University of Tennessee, Knoxville May 2023

B.Tech., Mathematics (First Class Honours), Ladoke Akintola University of Technology, Nigeria Feb. 2015
Thesis: *Mathematical modeling of a standard incidence epidemiological model with relapse*. Advisor: Dr. Samson Olaniyi

PUBLICATIONS

Peer-Reviewed Journal Articles

[1] **M. A. Lawal**, M. Demir, M. R. Kelly, and S. Lenhart (2026). An optimal control framework for multi-species harvesting in the southern Black Sea. *Natural Resource Modeling* 39:e70044. doi:10.1111/nrm.70044

[2] C. W. Chukwu, **M. A. Lawal**, D. Aldila, and G. Obaido (2026). Evaluating syphilis transmission among MSM population: a mathematical model for reinfection, two-stage treatment and treatment failure. *Frontiers in Public Health* 14:1841745. doi:10.3389/fpubh.2026.1841745

[3] S. Olaniyi, **M. A. Lawal**, and O. S. Obabiyi (2016). Stability and sensitivity analysis of a deterministic epidemiological model with pseudo-recovery. *IAENG International Journal of Applied Mathematics* 46(2):160–167.

Preprints

[1] **M. A. Lawal** (2026). State-level validation of syphilis and chlamydia transmission models in the United States: an open surveillance panel (2000–2024) and a held-out-year benchmark. *Preprints.org*. doi:10.20944/preprints202611.0141.v1

Under Review

[1] C. W. Chukwu, **M. A. Lawal**, K. R. Cheneke, and D. Wanduku. Simulation-based predictive modeling of behavioral heterogeneity in chlamydia transmission: insights from U.S. surveillance data.

In Preparation

[1] **M. A. Lawal**, C. Edholm, H. Correia, F. Agusto, S. Lenhart, W. Chukwu, D. Goldsman, F. Chirove, J. White, and S. Dobson. Modeling the interplay of vaccine hesitancy and disease dynamics.

DATA AND SOFTWARE

U.S. State Syphilis and Chlamydia Panel and Model Validation Benchmark, v1.0.0 Oct. 2026
Open panel of CDC-reported syphilis and chlamydia cases for all U.S. states, 2000–2024, with Census denominators, and a state-by-state held-out-year validation scorecard for two published STI transmission models. Panel: doi:10.5281/zenodo.23742262; scorecard: doi:10.5281/zenodo.23742264; code: github.com/maruf-lawal/us-sti-state-panel; preprint: doi:10.20944/preprints202611.0141.v1

AWARDS AND HONORS

- **Winner**, *Modeling the Invisible: Competition on Forecasting Viral Spread with Limited Data*, Georgia State University — June 2026
- University of Tennessee Open Publishing Support Fund Award ($1,800) — Sept. 2026
- UTK Mathematics Department Summer Dissertation Fellowship ($4,000) — June 2024
- Yueh-er, Hong-hsu and Clarence Cheng Kuo Fellowship, UTK Mathematics Honors Day ($1,000) — Apr. 2023
- UTK Mathematics Department Academic Excellence Fellowship ($1,660) — 2022
- UTK Mathematics Department Award ($2,000 in 2023; $2,300 in 2024) — 2023, 2024
- UTK Graduate School Fellowships ($2,000 in 2022, 2023, 2024, 2025; $1,249 in 2024; $1,018 in Aug. 2025 and Jan. 2026) — 2022–2026
- UTK Graduate School Award ($400) — 2022
- AMS Joint Mathematics Meetings Graduate Student Travel Grant ($1,430) — Jan. 2025
- AMS BEGIN Bootcamp Travel Grant ($300) — Jan. 2025
- UTK Graduate Student Senate Travel Awards ($650, $500, $800) — 2024–2026
- Gold Medal, National Mathematics Competition for University Students (NAMCUS), Abuja, Nigeria — Apr. 2015

PRESENTATIONS

Invited Talks

- Optimal sustainable management of multi-species harvesting in the southern Black Sea ecosystem. *Joint Mathematics Meetings*, Washington, DC — Jan. 2026
- Optimal sustainable management of horse mackerel harvesting in the southern Black Sea coast. *SIAM Southeastern Atlantic Section Meeting*, Knoxville, TN — Mar. 2025
- Optimal sustainable management of horse mackerel harvesting in the southern Black Sea coast. *Joint Mathematics Meetings*, Seattle, WA — Jan. 2025
- Modeling COVID-19 dynamics incorporating vaccine hesitancy. *103rd MAA Southeastern Sectional Meeting*, Knoxville, TN — Mar. 2024
- Modeling COVID-19 dynamics incorporating vaccine hesitancy. *Joint Mathematics Meetings*, San Francisco, CA — Jan. 2024

Contributed Talks

- Modeling the interplay of vaccine hesitancy and disease dynamics. *Modeling the Invisible Workshop*, Georgia State University, Atlanta, GA — June 2026
- Optimal sustainable management of horse mackerel harvesting in the southern Black Sea coast. *Mathematics Presentations Seminar*, UTK — Nov. 2024
- Modeling COVID-19 dynamics incorporating vaccine hesitancy. *Mathematics Presentations Seminar*, UTK — Sept. 2024

RESEARCH WORKSHOPS AND BOOT CAMPS

IMA Math-to-Industry Boot Camp (virtual) — June–July 2025
Team project: Proactive housing support—using public data to predict and prevent evictions.

IMSI Data Science Boot Camp (virtual) — June 2025

Mathematical Problems in Industry Workshop, Worcester Polytechnic Institute — June 2022
Team project: Patient data generation and machine learning on clinical data to predict heart-failure severity.

Graduate Student Mathematical Modeling Camp, University of Delaware — June 2022
Team project: Asymmetric division and modeling of interacting cell populations in the colonic crypt.

TEACHING EXPERIENCE

Positions

- Graduate Teaching Associate, University of Tennessee, Knoxville — 2022–present

- Graduate Teaching Assistant, University of Tennessee, Knoxville 2021–2022
- High School Mathematics Teacher, Learn-With-Ease Schools, Lagos, Nigeria 2019–2021
- Mathematics Tutor, Tractrain Consulting, Lagos, Nigeria 2018–2021
- Teaching Assistant, Federal Polytechnic Kaura Namoda, Zamfara, Nigeria 2015–2017

Instructor of Record

- Statistical Reasoning (MATH 115), UTK Spring, Fall 2026
- Basic Calculus (MATH 125), UTK (six semesters) 2022–2025
- Trigonometry and Analytic Geometry (MATH 122), Federal Polytechnic Kaura Namoda 2016

Co-Instructor

- Calculus I (MATH 141), UTK Fall 2025
- Calculus I-B Infused with Precalculus (MATH 132), UTK Fall 2025

Teaching Assistant

- Basic Calculus (MATH 125), UTK Spring 2022
- College Algebra (MATH 119), UTK Fall 2021

PEER REVIEW

Reviewer for the following journals:

- *Journal of Biological Dynamics*
- *PLOS One*
- *Scientific Reports*
- *Mathematical Modelling and Applications*
- *Mathematical Modelling and Numerical Simulation with Applications*

PROFESSIONAL SERVICE

- Reviewer, Undergraduate Poster Session, Joint Mathematics Meetings (San Francisco; Seattle) 2024, 2025
- Elected Senator, UTK Graduate Student Senate 2024–2025
- Graduate Student Representative, UTK Technology Fee Advisory Board 2024–2025
- Panelist, UTK SIAM Graduate Student Internship and Workshop Experiences Panel Oct. 2023

OTHER CONFERENCES ATTENDED

- Masamu Advanced Study Institute (MASI) Convening and Research Workshop, SAMSA (virtual) Nov. 2025
- One Health Student Leadership Summit, University of Tennessee, Knoxville Apr. 2025
- Society for Mathematical Biology Annual Meeting, Ohio State University, Columbus, OH July 2023
- Twelfth Annual MASI and Research Workshop, SAMSA Conference (virtual) Nov. 2022
- Eleventh Annual MASI and Research Workshop, SAMSA Conference (virtual) Nov. 2021

Meetings at which I presented are listed under Presentations.

SKILLS, AFFILIATIONS, AND TRAINING

Programming MATLAB, Python, R, Maple, LaTeX

Methods Deterministic and stochastic dynamical systems; optimal control theory; parameter estimation, identifiability, and uncertainty quantification; model calibration and validation against surveillance data; ensemble and probabilistic forecasting; statistical and machine learning methods

Affiliations AMS, SMB, SIAM, MAA, MIDAS, Nigerian Mathematical Society

Certifications Machine Learning with MATLAB, MathWorks (2024); Python for Data Science and Machine Learning, Advanced Python, and R for Data Science, LinkedIn Learning (2024); UTK Mathematics GTA Teaching Advising Program (2023)

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